Stress in doctors and dentists who teach

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Objective To explore the relationship between a teaching role and stress in doctors and dentists who teach.

Methods Medline, PubMed, BIDS database for social sciences literature, and the ERIC database for educational literature were searched using the key words 'stress' or 'burnout' with the terms doctor, physician, dentist, teacher, lecturer, academic staff, and university staff. Other books and journals known to the authors were also used.

Results Many studies have shown high levels of stress in doctors, dentists, teachers, and lecturers. A large number of factors are implicated, including low autonomy, work overload, and lack of congruence between power and responsibility. Doctors and dentists who take on a teaching role in addition to their clinical role may increase their levels of stress, but there is also evidence that this dual role may reduce job-related stress.

Conclusions Working as a doctor or dentist may entail higher levels of stress than are experienced by the general population. In some situations adding in the role of teacher reduces this stress, but more research is needed to explain this finding.

Keywords Adaptation, psychological; education, medical, *psychology; physician's role; stress, psychological, *teaching; review literature.

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Introduction

The jobs of doctors and dentists are inherently stressful, dealing as they do with people in pain or distress, while constrained by limited resources. Many doctors and dentists take on a dual role, having responsibility for teaching students and postgraduate trainees as well as caring for patients. Since the reforms of the undergraduate curriculum and of specialist training, the teaching and training responsibilities of consultants have become much more time-consuming, and may possibly contribute to stress.

The *NHS Plan*[1] suggests that the careers of consultants should be phased so that they carry different responsibilities at different stages. One of the responsibilities which could be removed from the many and delegated to a few consultants, perhaps later in their career, is the responsibility for teaching and training. Would such a separation relieve the strain caused by carrying too many roles? Or would losing the teaching and training role aggravate stress in doctors and dentists? We set out to review the literature on stress in doctors, dentists and teachers, to explore whether the stressors were similar and whether there was any evidence that taking on a teaching role aggravated or mitigated stress in doctors and dentists. We carried out a search on Medline, PubMed, BIDS database for social sciences literature, and the ERIC database for educational literature, using the key words 'stress' or 'burnout' with doctor/physician/dentist/teacher/lecturer/academic staff/university staff. We also used books, reports and papers already known to us.

Defining stress and burnout

Stress has been defined as a process which causes or precipitates individuals to believe they are unable to cope with the situation facing them, and the feelings of anxiety, tension, frustration, and anger which result from the recognition that they are failing in some way and the situation is getting out of their control.[2] Sustained stress may lead to anxiety and depressive reactions, as well as physical health problems such as coronary heart disease.[3] A well-validated measure of stress is the 12- or 28-question General Health Ques-

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Key learning points

Working as a doctor or dentist is stressful.

Working as a teacher or lecturer is stressful.

Doctors and dentists who also teach appear to have lower stress levels than those who do not teach.

More research is needed to explain this finding.

tionnaire (GHQ-12, GHQ-28), designed to pick up symptoms of psychological morbidity in populations.[4]

Burnout has been defined as a syndrome of emotional exhaustion, depersonalization (treating people in an unfeeling, impersonal way), and reduced personal accomplishment which occurs among individuals who work with people in some capacity. The most widely used measure for this is the Maslach Burnout Inventory.[5]

Stress in doctors and dentists

Many studies have shown high levels of stress in doctors, with psychological morbidity ranging from 19% to 47%,[6–9] compared with a rate of around 18% for the general employed population.[10]

A questionnaire study of 1133 consultants carried out by Ramirez *et al.* associated stress with work overload, followed by feeling poorly managed and resourced; managerial responsibilities; dealing with patients' suffering; keeping up to date; being responsible for the quality of work of other staff, and having to deal with relatives.[11] Those who felt insufficiently trained in management and communication expressed higher levels of depersonalization and lower levels of personal accomplishment than those who felt adequately trained. Job satisfaction was protective against job stress and depersonalization, with autonomy and feeling well managed and resourced acting as major contributors to job satisfaction.

An in-depth exploration of some of the factors causing stress was carried out by Allen *et al.* through a series of focus group discussions with hospital consultants.[12] The researchers found that the role of the consultant had become increasingly complex, with increasing demands from a variety of sources along with greater accountability to a variety of stakeholders. For example, Royal Colleges and postgraduate deans were demanding ever higher standards of trainee supervision, while the service contribution of trainees was steadily eroded by reductions in their working hours and duration of training. These responsibilities had been imposed on consultants who felt decreasing power to influence the means by which they could best be handled. The consultants felt themselves to be ultimately accountable in terms of patient care but without the power to determine how these services were to be delivered. The greatest rewards in terms of job satisfaction were provided by patients but they also caused some of the greatest stress in that they were increasingly demanding of time and energy, had unrealistic expectations, were increasingly aggressive and, perhaps most stressful of all, were more and more likely to complain. Lack of congruence between power and responsibility was again an important theme, which is echoed in other reports showing that lack of autonomy, lack of control over work, and reduced participation in decision-making are important factors causing stress and anxiety in doctors.[13–15]

In studies of general practitioners, the four stressors most predictive of job dissatisfaction and stress were the demands of the job and patients' expectations; interference with family life; constant interruption at work and home, and practice administration.[16,17]

Dentists have similar levels of stress and depression.[18] In an interview study, the factors which dental practitioners referred to most often as causing stress were system changes in running a dental practice, and rising patient expectations. Staff turnover, financial worries, a feeling of having to see too many patients, aggression from patients, litigation and the stress of working with other team members were also cited.[19] Working with colleagues may be a source of stress, but so is isolation.[20]

Stress, burnout, and depression must be a matter for concern in professions where humanity, compassion and competence are such essential attributes.

More information about some of these studies is shown in Table 1.

Stress in teachers and lecturers

Levels of stress and depression in school teachers have been shown to be as high or higher than those in doctors and dentists.[21] Many of the stressors are similar: workload, lack of resources, poor relationships with colleagues, unrealistic expectations from managers and difficult interactions with pupils and parents.[22]

A study of 1110 school teachers in Canada showed correlations between isolation and occupational stress.[23] Another large Canadian study found a relationship between bureaucratic interference and emotional exhaustion; this in turn was related to greater depersonalization, feelings of lower personal accomplishment, and greater somatization.[24] Other studies have shown stress in teachers to be related to working hours,[25] role overload,[26] and concerns such as pupil misbehaviour, time⁄resource difficulties, professional recognition needs, workload and poor relationships.[27]

Stressors in higher education, including medical and dental education, where students are more likely to have made a positive decision to attend and teachers are likely to have competing demands in terms of academic productivity, differ from those in school level education. Explorations of the roles and stress in academic life have been extensively described by Fisher, who noted that university staff are expected to teach, meet tutorial, laboratory or seminar commitments and at the same time carry out research, run experiments, obtain research funding, and write papers and books.[28] Academic staff have traditionally enjoyed a high degree of personal control over their jobs, but this has diminished in recent years. The development of the research assessment exercise in the United Kingdom, which links university income with research achievements, creates significant pressure to perform research and seek funds with which to do this, at the expense of teaching. In addition to academic demands from their employers, there are increasing demands from students, whose views of what constitutes good performance may differ from those of their teaching staff.[29]

Teaching has been subjected to a prolonged period of change in recent years. Teachers have been required to assimilate changes, modify their behaviour and evaluate success by assessing the progress of their pupils. Travers & Cooper surveyed a random sample of 1790 teachers drawn from a cross-section of school types, sectors and teaching grades across the UK.[30] They found that this group of teachers reported more manifestations of stress than comparable occupational groups, including general practitioners and dentists. The top ten sources of pressure for this group were lack of support from the government; constant changes taking place within the profession; lack of information as to how the changes were to be made; society's diminishing respect for the profession; the move toward a national curriculum; salary out of proportion to

Table 1 Selected papers on stress in doctors and dentists

Reference number	Setting	Study instrument if specified	Sample size	Findings
8	UK teaching hospitals	GHQ	445 consultants 93 house officers	Consultants had higher levels of stress than junior doctors
9	UK hospitals	GHQ, SCLDS	238 house officers	Emotional disturbance in 50% and evidence of depression in 28% Overwork was the most stressful aspect of their jobs
11	UK hospitals	GHQ, MBI	1133 consultants	Job satisfaction, and provision of training in communication and management skills, protect consultants' mental health
16	UK hospitals and general practice	GHQ, HADS	81 hospital consultants 322 general practitioners 121 senior hospital managers	47% of respondents scored positively on GHQ GPs more likely to be depressed than managers, and more likely to show suicidal thoughts
18	UK general dental practices	In-depth interviews and GHQ	20 NHS and 30 independent dentists	Time pressure the most frequently identified stressor No significant differences in stress between the two groups, but some indications that private practice was less stressful
19	UK general dental practices	In-depth interviews	10 dentists	Uncertainty about the future organization of dental care was the most important new work pressure
20	UK general dental practices	MBI	325 dentists	Burnout was related to features of work organization structure. Higher numbers of dentists in a practice appeared to have a protective effect against some aspects of burnout

GHQ, General Health Questionnaire; SCLDS, Symptom Check List Depression Scale; MBI, Maslach Burnout Inventory; HADS, Hospital Anxiety and Depression Scale.

workload; having to produce assessments of pupils; dealing with basic behavioural problems; lack of non-contact time, and the fact of being a good teacher not necessarily meaning promotion. The two major predictors of job dissatisfaction were the management structure of the school and lack of status or promotion. The NHS doctor or dentist with a teaching or training role would recognise many of the same factors.

Stress in doctors and dentists who teach

If high levels of stress are a feature of both health care and teaching, then are stress levels even higher in those who combine the roles? Harden has suggested that there are particular pressures on medical teachers which involve changes in health care delivery, in public expectations and in medical education.[31] He has drawn together the interacting factors causing stress in medical teachers in the form of a complex equation. The stressors involved in the equation include the teaching workload, the number of teaching weeks, the number of students, the intensity of teaching, the tasks to be undertaken by the teacher and the number of staff; additional factors included the clinical workload, research activities and administrative workload. The equation also incorporates other factors, such as loss of teacher autonomy, role conflict, rewards and job satisfaction, available resources and forms of support for the teacher. In particular, increasing demands are being made on the teacher in medicine with regard to student assessment.

Postgraduate training for junior doctors has recently been reformed adding to the pressures on their consultants.[32,33] The reforms were intended to shorten the time it took to train consultants in the UK, by improving the structure and content of training programmes. They were intended to be accompanied by an increase in the number of consultants, but this increase has not so far materialized. The reforms have led to more work; further loss of autonomy for consultants, who can no longer appoint their own trainees, and the threat of losing recognition for training if externally imposed standards are not met.[12] The introduction of a formal 'record of in-training assessment' has increased the burden on consultant trainers. They must provide objective reports about the trainee's progress, but writing an adverse report and discussing it with the trainee can be stressful. Further, the trainer who agrees to take on a trainee who has been identified as poorly performing carries a heavy burden of responsibility to his or her peers and patients, as well as to the trainee.[34]

Despite all this, we could not find any research which demonstrated that doctors and dentists with undergraduate or postgraduate teaching responsibilities were any more stressed than their peers.

Factors which reduce stress

Teachers who take a course in human relations may suffer less stress.[35] Hall *et al.* studied 32 male and female teachers and found that following such a course there was a reduction in reported stress, indications of a more humanistic orientation towards pupil control, and an increased sense of an internal locus of control.

An American study reported on subjects recruited from a University teaching programme who were assigned to meditation training or a control group.[36] The meditation group subjects were found to have significantly reduced stress symptoms compared with those in the control group.

In Sweden, a stress control programme was targeted at high school teachers.[37] A treatment group of 56 teachers followed the programme, which involved discussions about various aspects of stress. Compared with a control group of 33 teachers, the programme participants reported significantly less stress following the programme.

Senior faculty members in a Californian college were invited to an off-campus retreat to identify policies and practices which served as a bridge or barrier to their own professional development.[38] Following these meetings, specific action was taken to create projects to bring lecturers to campus to assist tenured female faculty members in completing their doctoral dissertations, and to restructure a compulsory beginning of term faculty meeting to include discussion of educationally related topics.

A study of black women lecturers in Botswana[39] identified the role of positive thinking, a form of cognitive therapy, as a stress treatment measure.

Although excessive workload was a constant theme in studies of stress in teachers and doctors, effective interventions may involve extra work or responsibilities. Chinese secondary school teachers[40] found that teachers who took on additional pastoral duties felt a greater sense of accomplishment than others, despite having to work harder.

Successful techniques for reducing stress in dentists included 'switching off' from dentistry and taking exercise.[18] General practitioners in partnerships were less depressed than those who worked in single-handed practices, and those who had half-days free had significantly lower depression scores than those who did not.[41] A review of research on stressed doctors highlighted the importance of protecting time for practitioners' own professional development, informing employers if working conditions or hours were unacceptably stressful, and recognising the benefits of interprofessional working.[42]

Can a teaching role mitigate stress in doctors and dentists?

There is evidence that the teaching role may reduce stress. A recent survey of stressors affecting psychiatrists revealed that teaching medical students and doctors caused them little or no stress.[43] In a survey of general practitioners, the degree of involvement in training was inversely related to depression scores. There was also a significant association between low anxiety scores and being involved in training.[41]

Why should a teaching role mitigate stress in doctors? One factor may be professional status and peer recognition. Wright *et al.* surveyed trainees at four teaching hospitals in the USA and asked them to name physicians whom they considered to be excellent role models.[44] Five attributes were found to be independently associated with being an excellent role model: spending more than 25% of time teaching; spending 25 hours or more per week teaching, and conducting rounds during time spent as an attending physician; stressing the importance of the doctor–patient relationship during teaching; teaching psychosocial aspects of medicine, and having served as a chief resident. Promotion prospects may be improved by an active teaching role. In a study of factors associated with promotion of academic staff within medical schools in the USA and Canada, teaching skills, clinical skills and mentoring skills were the three most highly ranked.[45] The authors concluded that most, but not all, promotion committees assign high importance to the special contributions of clinician-educators.

Frederick Herzberg highlighted the value of job enrichment in improving job satisfaction and motivation.[46] He suggested that, paradoxically, adding new responsibilities could motivate employees to work harder, if as a consequence they gained more control over their environment.

For professionals motivated by the desire to 'make a difference', training young doctors to treat a condition seems more productive than simply doing the treating oneself.[47] In the UK, medical consultants who were also clinical tutors or directors of education, and hence significantly involved with the management of medical education, were more positive about their work than other consultants. These doctors had their own place of retreat, their own budgets, their own staff and their own specific roles, and these led to a sense of control and autonomy despite the increased work involved.[12]

Discussion

We found no evidence in our review of the literature that doctors and dentists with teaching or training responsibilities were any more stressed than their non-teaching counterparts. On the contrary, there was some evidence that a teaching or training role might mitigate stress. The reasons for this must be speculative, but might include lessened isolation, increased self-esteem in response to the attention of students, a sense of autonomy over what and when to teach, power over those in a more junior position, added interest in patients as a source of teaching opportunities, and a sense of helping the future patients of those being taught.

The evidence for the impact of a teaching role on job-related stress is confusing. Teaching and academic work involve exposure to stresses such as potentially reduced status; increased external policing (e.g. by academic assessors), and thus reduced autonomy, and altered interpersonal relationships with colleagues such as trainees. There is, however, a more positive side to the teaching role, which may also be associated with increased status, more autonomy, and greater occupational variety and interest. It seems likely that teaching does not influence job-related stress *per se*, rather it is the personal and professional circumstances affecting that teacher which are of most importance. There are, for example, enormous differences in both the clinical and the teaching role between a teaching hospital consultant, a general practice trainer, and a junior doctor providing *ad hoc* teaching for his or her colleagues on the ward.

Much of the evidence cited in this review can be criticised on several grounds. Sample sizes are small in many of the studies, some of which also have low response rates, and a number of papers demonstrate clear selection bias. Chambers' paper, for example, compares the health and lifestyles of GPs and teachers, but it is based on small sample sizes and low response rates, which reduces the reliability of its conclusions.[21] The generalizability of much of the published data can also be questioned, given the narrowness of the selection criteria shown in several of the papers mentioned. McManus *et al.* have criticised the research on stress in doctors on the grounds that published papers may exaggerate stress as a consequence of questionnaires that specifically focused on stress, thus creating a demand characteristic.[48] Their research using the GHQ-12 demonstrates lower levels of stress among doctors than those found in other studies, and which are equivalent to levels in the general population, but they acknowledge that the comparison can only be approximate because of a range of confounding factors.

Conclusions

This review of current literature reveals a number of gaps in understanding of the factors involved in occupational stress in doctors and dentists, and the association between these factors and a teaching role. Further research is required to tease out these links, identify both positive and negative factors implicated in job-related stress, clarify any causal relationships that may exist, and accurately establish the prevalence of stress within these professions. The relationship between a teaching role and job-related stress remains poorly understood, and research is needed to disentangle the general from the specific factors. Teachers at certain medical and dental schools may, for example, experience much higher or lower job-related stress than those at others, as a result of particular cultural or organizational circumstances, and these need to be separated out from generic factors applicable to all such establishments. Comparison should also be made across different teaching and learning environments, from medical and dental schools to hospitals and primary care practices. The time is long overdue for moving away from a professional culture that accepts and expects job-related stress as an inevitability, and to concentrate on improving the working conditions of all in the profession.

Contributors

JH and EP performed the original literature review and wrote the first draft. HR followed up the literature review and wrote the final version of the paper.

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